INCH-POUND

MIL-L-53111(ME)
12 September 1991

MILITARY SPECIFICATION

LABORATORY, PETROLEUM, SUPPLEMENTAL, SEMITRAILER MOUNTED

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the supplemental semitrailer mounted petroleum laboratory fully equipped with testing equipment, chemicals, and supplies to analyze fuels, lubricating oils, greases, and hydraulic fluids in a theater of operations for 60 days.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

W-C-375	- Circuit Breakers, Molded Case, Branch Circuit and Service.
W-P-115	- Panel, Power Distribution.
W-R-550	- Rod, Ground (with Attachments).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6640

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TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer Resisting.
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-1055	- Barrier Material, Waterproof, Flexible.
PPP-C-1752	- Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible.
PPP-T-60	- Tape: Packaging, Waterproof.

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MIL-P-116	- Preservation, Methods of.
MIL-P-514	- Plate, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-B-7883	- Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-F-16377	- Fixture, Lighting, Incandescent, General Lighting 100 Watts, 120 Volts, Drip-Proof 15 Degrees, Symbols 89 and 90.2.
MIL-F-16377/12	- Fixtures, Lighting; Fluorescent, General Lighting, 60 Watts, 120 Volts, 60 Hertz, Watertight, Symbols 333.1, 333.2, 341.1, 341.2, 342.2 and 342.3.
MIL-C-22992	- Connector, Plugs and Receptacles, Electric, Waterproof Quick Disconnect, Heavy Duty Type.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-S-46756	- Semitrailer Van: 10 Ton, 4 Wheel, Series.
MIL-C-53039	- Coating, Aliphatic Polyurethane, Single Components, Chemical Agent Resistant.
MIL-V-62038	- Vehicle, Wheeled, Preparation for Shipment and Storage.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standard for Federal Services.
FED-STD-313	- Material Safety Data Sheets, Preparation and the Submission of.

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MIL-STD-101	- Color Code for Pipelines and for Compressed Gas Cylinders.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-209	- Military Standard Slinging and Tiedown Provisions for Lifting and Tying Down of Military Equipment.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.

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MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1366	- Materiel Transportation System Dimensional and Weight Constraints, Definition of.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiels.
MIL-STD-1791	- Designing for Internal Aerial Delivery in Fixed Wing Aircraft.
MIL-STD-1857	- Grounding, Bonding and Shielding Practices.
MS35000	- Battery, Storage, Lead-Acid, Waterproof.
MS52269	- Semitrailer Van, Petroleum Laboratory, 10 Ton, 4 Wheel, M822.

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

13217E1061	- Rivet and Riveting Requirements.

COMPONENTS PARTS LIST

90106	- Supplemental, Semitrailer Mounted Petroleum Laboratory.

(Copies of drawings and components parts list required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FSP, Fort Belvoir, VA 22060-5606.)

DEPARTMENT OF LABOR (DOL)

29CFR1910	- Occupational Safety and Health Administration (OSHA).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

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DEPARTMENT OF TRANSPORTATION (DOT)

49CFR171-190 - DOT Hazardous Materials Regulations.
49CFR393 - DOT Federal Motor Carrier Safety Regulations (FMCSR).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

UNITED STATES ARMY ENVIRONMENTAL HYGIENE AGENCY (USAEHA)

Technical Guide -030, Nov 1978, Guidelines on Design of Chemical Laboratory Hoods.

(Application for copies should be addressed to the United States Army Environmental Hygiene Agency, HSE-OI/WP, Aberdeen Proving Ground, MD 21010-5422.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

C73.11 - Dimensions of Attachment Plugs and Receptacles, General Purpose Type, 125 Volt Plugs and Receptacles, 2 Pole, 2 Wire, Grounding.
Y14.6 - Metric Supplement.
Z358.1 - Emergency Eyewash and Shower Equipment.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standards

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding Code - Steel.
D1.2 - Structural Welding Code - Aluminum.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

501C - Mobile Vehicle Standards.
70 - National Electric Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J429 - Mechanical and Material Requirements for Externally Threaded Fasteners.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The supplemental semitrailer mounted petroleum laboratory (hereinafter called supplemental laboratory) shall be in accordance with the parameters and design limitations specified herein. The supplemental laboratory shall be compatible with an M818, 5 ton, 6x6 truck tractor, (or equal). The supplemental laboratory shall be provided a catwalk (NSN 2510-01-205-5990) that will attach to the rear platform of each trailer, coupling the units together.

3.2 First article. Unless otherwise specified (6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. The material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of the specification.

3.3.1 Material deterioration prevention and control. The supplemental laboratory shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable

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operation and storage environments to which the supplemental laboratory may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-T-704 and MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish, or treatment for use with the component and subcomponent and shall make this information available, upon request, to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, except electrical and electronic components, pieces, and parts incorporated in the supplemental laboratory may be newly fabricated from recovered materials to the maximum extent practicable, provided the supplemental laboratory produced meets all other requirements of this specification. Used, rebuilt, or remanufactured components, pieces, and parts shall not be incorporated in the supplemental laboratory.

3.4 Design. The supplemental laboratory shall be designed within the parameters and limitations herein. The design shall be such as to limit or prevent conditions hazardous to personnel or deleterious to equipment. The supplemental laboratory shall comply with the Occupational Safety and Health Administration (OSHA) standards in 29CFR1910 and the utility systems shall comply with the mobile vehicle standards in NFPA-501C. In addition, the 120/208 volt ac electrical system shall be in accordance with the National Electric Code (NEC), NFPA-70. Any data (e.g., shop and fabrication drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a supplier to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or designated representative.

3.4.1 Semitrailer. Unless otherwise specified (see 6.2), the semitrailer shall be in accordance with MIL-S-46756 and MS52269.

3.4.2 Semitrailer design modifications. The semitrailer (van) shall be divided into three compartments. Each compartment shall be separated from the next by a bulkhead. Each bulkhead shall be of metal construction and shall be designed so as to reduce the noise transmission between compartments. Both the rear and forward bulkheads shall be of modular construction and removable. The rearward bulkhead shall incorporate a swinging door with the ability to be latched and secured during transit. The door minimum opening shall be 24 inches wide by 76 inches high. The door shall be centered within the bulkhead. The door shall contain a window made of polycarbonate sheet, 8 inches high minimum by 10 inches wide minimum.

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3.4.2.1 Floor design. The supplemental laboratory floor shall be of laminar design. The floor shall consist of structural aluminum sheet, polystyrene foam and a marine plywood subfloor with beige colored linoleum over it. Structural members between the aluminum sheet and subfloor shall be on 16-inch centers and shall run the length of the semitrailer to provide load support.

3.4.2.2 Exterior wall modifications. The roadside exterior wall shall contain two fume exhaust vent ports and an external power connection. One fume exhaust vent shall be located adjacent to the forward bulkhead and one fume exhaust vent shall be located adjacent to the rear bulkhead. The external power connection shall be located forward, above the step in the semitrailer frame (see figure 1). The curbside exterior wall shall contain the water inlet and outlet ports (see figures 1 and 2). The forward curbside and roadside walls shall contain air inlets for purge and ventilation air make-up. Actuated purge exhaust vents shall be contained within both the roadside and curbside walls, one each.

3.4.2.3 Ceiling design. The environmental plenum and lighting fixtures shall be integral with the ceiling design. The environmental plenum shall form the ceiling center section. Hinged panels containing the light fixtures shall be directly adjacent to the environmental plenum. The environmental plenum shall run the entire length of the semitrailer, be constructed of 0.06 stainless steel sheet, and be insulated for the entire length. The hinged panels shall run from the forward bulkhead to the rear bulkhead and from the rear bulkhead to the rear wall of the semitrailer. Curbside and roadside ceiling panels shall run from the forward bulkhead to the rear wall of the semitrailer. Acoustic treatment shall be provided to meet the noise requirements of 3.19.

3.4.2.4 Interior wall design. The interior wall design shall consist of 0.09-inch stainless steel panels run longitudinally along the roadside and curbside interior wall structure.

3.4.2.5 Cabinetry. The main and aft compartments shall have 1-inch thick solid epoxy countertops, with steel cabinetry below, on the roadside and curbside wall areas. The countertops shall serve as mounting surfaces for major items of equipment (see figures 2 and 3). Support apparatus and portable equipment shall be located within the cabinets. The cabinetry shall not block, or interfere with, the aisle or means of egress from any compartment. The cabinet configuration shall conform to the specifications contained herein. The cabinet height shall be in accordance with the requirements of 3.18.

3.4.2.6 Undercarriage modifications. Roadside and curbside undercarriage compartments shall be provided for stowage of miscellaneous gear (e.g., cable reel, hose, 12-volt dc batteries, access ladders, etc.). The undercarriage shall provide for the mounting of a waste water tank. These stowage compartments, and wastewater tank if necessary, shall be removable for C-141 or larger air transport.

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3.5 Components. Items of equipment shall be as specified in the components parts list (CPL) 90106. When specified (see 6.2), the items of equipment shall be furnished by the contractor and shall be located in the supplemental laboratory as specified herein (see figures 2 and 3). That equipment which is countertop mounted shall be secured in such a fashion as to ensure transportability while maintaining operational character. The equipment to be mounted on the countertop should be positioned as specified in figures 2 and 3.

3.5.1 Fume hood. The fume hood for the supplemental, semi-trailer laboratory shall incorporate a hood exhaust fan with the capacity to provide 100 feet per minute (fpm) average velocity across the face of the hood with the sash at the fully open position. The hood shall have an independent exhaust duct and shall be airtight so that no fumes can circulate back into the laboratory. Furthermore, the fume hood shall be in compliance with the following documents:

a. Technical Guide (TG)-030, November 1978, Guidelines on Design of Chemical Laboratory Hoods. USAEHA.
b. DOL 29CFR1910. OSHA.

3.6 Laboratory purging. Supplemental laboratory purging shall be accomplished by operating the evaporator coil fans within each air conditioning unit with all electrical power to the laboratory interior suspended during the purge cycle. During purge, dampers in the air conditioning intake filter box shall allow only external air to be supplied to the fans. Electrical componentry from the purge control to the environmental control units shall be explosion proof. Exhaust vents shall be located low on the roadside and curbside walls to ensure evacuation of heavier-than-air gases or vapors. Purge system components shall be selected to accomplish a purge of four air exchanges in no more than 5 minutes. A purge cycle timer shall be provided on the exterior wall near the door and shall be adjustable from 5 minutes to 20 minutes.

3.6.1 Purging on deployment. Purging on deployment shall be automatic. No electrical apparatus shall be energized within the confines of the supplemental laboratory during purge. No entry of personnel shall be required prior to, or during, the purge cycle.

3.6.2 Purging during employment. Purging during employment shall be automatic. Power will be automatically suspended when a level of flammable fumes in excess of 40 percent is detected by the fume detection system. The cycle shall be the same as on deployment (5 minutes). Automatic purging will continue in 5-minute segments until the level of flammable fumes drops below 40 percent or the vehicle is completely disconnected from its power source.

3.6.3 Purging override. Purging override is accomplished automatically for no more than 60 minutes. When power to the supplemental laboratory is suspended for any reason, the supplemental laboratory shall return to

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operational status without going through a purge cycle if the power loss is for less than 60 minutes.

3.7 Interior environment. The semitrailer interior environment shall be controlled during operation. The interior environment shall be in accordance with 3.18 except as noted herein.

3.7.1 Ambient temperature. Heating shall be provided to maintain interior dry bulb temperature above 50 °F. Air conditioning shall be provided to maintain an interior effective temperature at or below 75 °F.

3.7.2 Air supply. Ventilation air shall be introduced at a minimum of 30 sc feet per minute; approximately two thirds shall be external air. The supply of external air shall be variable from 5 percent to 100 percent during heating and cooling of the trailer by means of controlled damper vents on the air conditioning intake plenum.

3.7.3 Air conditioning units. Four modified environmental control units (ECU's) (NSN 4120-00-168-1781 or equal) shall be installed on the front wall, exterior of the semitrailer. The air conditioning units shall not transfer vibration to the semitrailer. No air leaks shall be present at the intake or return interface. The air conditioning units shall be grounded. Piping shall be provided to remove the total condensate from the air conditioning units. Each air conditioning unit shall be wired to allow the evaporator fans to run during purge.

3.7.4 Ducting. Insulated stainless steel air conditioning ducting shall be provided to service the general environment. The stainless steel sheet used in fabrication shall be not less than 0.040 inch and not more than 0.048 inch thick.

3.7.4.1 Environmental plenum. A general environment plenum shall run from the environmental control units rearward, forming the center ceiling panel, through the forward and rear bulkhead and terminate at the rear semitrailer wall. Adjustable vents shall be installed in the duct to provide for balancing the air flow.

3.7.5 Heating. Heating shall be accomplished by the use of heating elements within the environmental control units.

3.7.6 Environmental controls. The environmental controls shall be located in one panel/switchbox positioned in the main laboratory compartment, on the roadside interior wall. The controls shall be as specified in 3.18 and labeled in accordance with 3.18.

3.8 Utility systems. The supplemental laboratory shall provide self-contained water, compressed air, and vacuum utility systems. The apparatus shall be located in the utility compartment and each service shall be distributed as required via copper or stainless steel tubing.

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3.8.1 Water system. The water system shall supply clean, deionized water to the eye wash/safety shower, sink, and to outlet valves located within the service shelves located above the cabinetry. The system shall include, but not be limited to, water tank (60 gallon capacity minimum), pump, filter set, wall mounted deionizer set, gate valve(s), check valve(s), pressure regulator, quick-disconnect inlet (fill pipe) and outlet (drain overboard), and waste water holding tank. The water system's supply and drain lines shall be separate. The water system shall be able to use either independent onboard or facility dependent water supplies. A regulator shall be installed to prevent overpressurization of the supply lines. The water drain system, originating at the sink, shall capture the wastewater in a tank located on the undercarriage for later disposal. This wastewater tank(s) shall have a capacity of at least 120 gallons. The refuse water tank shall incorporate three thermostatically controlled, 1500-watt, strip heaters to prevent freezing. Compression or flared fittings shall connect all tubing.

3.8.2 Compressed air. The compressed air system shall supply pressurized, oil free air at 35-85 psi to outlets contained within the service shelves located above the roadside and curbside cabinetry. An external quick-disconnect fitting shall be provided to allow pressurized air to be used outside of the supplemental laboratory. The air compressor shall be of the electric motor driven compressor type with reservoir tank in accordance with American Society of Mechanical Engineers (ASME) requirements.

3.8.3 Vacuum. The vacuum system shall supply vacuum to outlets contained within the service shelving located above the roadside and curbside cabinetry.

3.8.4 Color code marking. All utility piping and valves shall be color coded in accordance with MIL-STD-101.

3.9 Electrically operated utility and testing equipment and systems. The electrically operated utility and testing equipment and systems shall be as specified and shall operate as intended without malfunction or evidence of binding or overheating.

3.9.1 Primary electrical system. The primary electrical system shall be 3 phase, 120/208 volt ac, 5-wire, 60 Hz, 100 amperes. An uninterruptible power supply (UPS) shall be utilized to provide the voltage requirements of the computer and its peripherals. The primary electrical systems shall provide power to the 4 air conditioning units, the purging fan/blowers, and laboratory equipment requiring 120/208 volts.

3.9.2 Secondary electrical system. The secondary electrical system shall provide power for the lighting systems, wastewater tank heaters, and the laboratory devices requiring 110 volts ac and shall be derived from the 120/208, 3-phase, 5-wire supply with consideration for load balancing.

3.9.3 Dedicated power. The power required by the computer and its peripherals shall be filtered, shielded, and buffered in accordance with

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MIL-STD-461. No additional power taps shall be available on the computer line other than those required by the data system. The line shall run from the power regulator directly, with no interruptions, to the computer and its peripherals.

3.9.4 Power cable and power entry receptacle. A power entry receptacle shall be located on the forward roadside van exterior. A 24-volt dc-powered cable reel assembly shall be provided on the undercarriage of the van for a 100-foot cable. The main power receptacle and its mating cable shall conform to MIL-C-22992. The receptacle shall be class L, shell size 44, for minimum 100-ampere current rated service, 5-wire, 3-phase, 50-60 Hz ac. The receptacle shell shall be oriented to shell master key position no. 4 (at 120 degrees for 120/208 volts ac use). The main power receptacle and cable assemblies shall be rated weatherproof in accordance with NEMA standards in the National Electric Code. The receptacle timer shall be labeled "POWER ENTRY," 120/208 volts ac, 50/60 Hz, 3-phase.

3.9.5 Emergency electrical system. The van shall have a 24-volt dc emergency electrical system. The exterior lights (i.e., taillights, brake lights, clearance lights) shall be compatible with the towing vehicle when 12-volt lamps are used. Each of the van's three compartments shall contain one 24-volt emergency lighting fixture. The batteries supplying the 24 volts required for the system, two 12-volt MS35000 size 6TN, shall be located in the cable reel box. A battery charger shall also be provided to maintain battery charge.

3.9.6 Circuit panels. The power distribution panels shall be in accordance with W-P-115, type 1, of the dead front design. Circuit breakers shall be in accordance with W-C-375.

3.9.7 Primary circuit panel. The primary circuit panel shall be located on the roadside interior bulkhead dividing the utility and the main laboratory compartments.

3.9.8 Instrument and support devices. Instrument and support devices which are permanently affixed to countertops or anchored in place shall be "hard wired" through switchboxes or motor starter-relay boxes mounted on the curbside or roadside walls, as applicable.

3.9.9 115-Vac outlets. Duplex 115-Vac outlets shall be provided above the counter surfaces within the service shelves. All outlets shall be "one-way" 120 volts ac type in accordance with ANSI C73.11, UL approved, 3-wire. The outlets shall be labeled to indicate voltage, frequency, and phase characteristics.

3.9.10 208-volt receptacles. 208-volt receptacles shall not be available in either the laboratory or the aft compartment.

3.9.11 Fume hood exhaust fan controls. Fume hood exhaust fan/blower(s) shall be controlled by switchboxes mounted in the fume hood service area.

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3.9.12 Light switch and lamp location. Light switch and lamp location shall be as specified in accordance with the minimum illumination requirements in accordance with 3.18.

3.9.13 Blackout lamps. Blackout lamps shall be integral with the fluorescent light fixtures in accordance with MIL-F-16377 and MIL-F-16377/12.

3.9.14 Microswitch installation. A microswitch shall be installed in each of the van access doors. Each compartment shall be provided with an override switch to bypass the microswitch circuit. Under normal conditions, opening any access door shall activate the blackout lamp(s) and deactivate the incandescent and fluorescent lighting in the compartment(s) serviced by that door.

3.10 Electrical wiring. All electrical wiring shall comply with the National Electrical Code (NEC) of the National Fire Protection Association. Cables and wires shall be in accordance with 3.18.

3.10.1 Cords. All electrical cords shall be as short as practicable while allowing for maintenance.

3.10.2 Permanent wiring. Major components of electrical equipment shall be tied into the electrical system permanently and shall be equipped with "ON" and "OFF" switches as well as circuit breakers or fuses.

3.10.3 Protection. Grommets shall be used to prevent chafing, rubbing, and abrading where wires, harnesses, or cables pass through any structural member or bulkhead. Cushioned straps shall be used to support wiring, harnesses, and cables. The maximum unsupported length of any wire, harness, or cable shall not exceed 36 inches. Wiring and cabling run through conduit need not comply for that length captured by the conduit. Mechanically mounted cable clamps shall be provided to ensure correct routing of electrical cables within and between equipment items to ensure that cables do not hinder or obstruct equipment maintenance and to prevent chaffing due to contact with adjacent structures.

3.10.4 System grounding. All electrical equipment and the semitrailer van shall be grounded in accordance with NEC and OSHA requirements as well as MIL-STD-1857. Ground busses and frames shall be grounded to each other and to a main grounding terminal in or near the exterior main power receptacle on the front road side. A ground rod shall be furnished in accordance with W-R-550, type II, class B. Stowage provisions shall be provided within the underslung semitrailer overpack storage on the roadside. Grounding instructions shall be furnished on a metal plate near the exterior power receptacle. Vehicle ground and neutral shall be separate.

3.11 Safety equipment. The process definition of the supplemental laboratory indicates that certain tests may endanger the operating personnel if the laboratory does not incorporate safety devices and equipment in its

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design. The equipment shall be positioned as specified to ensure the safest working areas while still enabling the laboratory to fulfill its mission.

3.11.1 Fume hood extinguisher. The fume hoods shall each contain an automatic fire extinguisher, type ABC dry chemical, with a fusible link-actuated head.

3.11.2 Wall-mounted fire extinguishers. Four wall-mounted, portable, bromochlorofluoromethane fire extinguishers shall be provided: one 2-3/4-pound unit located in the aft compartment; one 2-3/4-pound unit attached to the fume hood; one 2-3/4-pound unit positioned on the curbside wall next to the curbside access door of the main compartment; and one 5 pound unit positioned vertically on the floor to the left of the utility compartment access door.

3.11.3 First aid kits. Two wall-mounted first aid kits shall be provided: one specialty burn first aid kit shall be located above and to the right of the sink in the main compartment, and one general first aid kit shall be located in the rear compartment on the roadside half of the aft bulkhead.

3.11.4 Eye wash/safety shower. An eye wash/safety shower, hand-held type, shall be provided on the roadside wall countertop at the sink location. The eyewash/shower shall meet the requirements of ANSI Z358.1 and have sufficient hose length to provide complete mobility. Control valves shall be designed so that flow remains on without use of the operator's hands.

3.11.5 Gas detector. A gas detector, toxic and petrochemical, shall be located in the main compartment on the curbside wall, adjacent to the rear bulkhead.

3.11.6 Flammable liquid storage. A flammable liquid storage cabinet shall be provided on the roadside wall under the fume hood.

3.12 Safety equipment marking. Flammable liquids shall be stored in cabinets marked with a red (#28913) cross and warning labels. Safety and personnel equipment shall be stored in cabinets painted with a green (#14187) cross. All safety equipment shall be clearly marked in accordance with 3.18.

3.13 Placards. Major processing equipment shall be identified by placards indicative of any hazard to operating personnel in accordance with 3.18.

3.14 Labels. Warning labels shall be placed conspicuously in areas which may be hazardous to personnel as dictated by the testing processes (i.e., heat, high voltage, toxic fumes or vapors, flammable storage, etc.) in accordance with 3.18.

3.15 Receptacles, fixtures, and access panels. All electrical receptacles, fixtures, and access panels shall be marked in accordance with MIL-STD-454.

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3.16 Utility lines. Utility lines - "WATER", "COMPRESSED AIR", and "VACUUM" - shall be labeled clearly and unambiguously as to content, pressure, heat, cold, or other hazardous property in accordance with 3.18.

3.17 Illumination. Illumination of all work spaces shall be in accordance with 3.18.

3.18 Human factors engineering. The supplemental laboratory shall comply with the design criteria of MIL-STD-1472. Special design emphasis shall be given, but not limited to general requirements (4), control/display integration (5.1), visual displays (5.2), controls (5.4), labeling (5.5), anthropometry (5.6), workspace design (5.7), environment (5.8), design for maintainability (5.9), operational ground/shipboard vehicles (5.12), and hazards and safety (5.13), as applicable.

3.19 Noise limits. The supplemental laboratory shall conform with MIL-STD-1474 requirements, with the exception of MIL-STD-1474 paragraphs 5.2, 5.3, and 5.4, when tested in accordance with 4.5.10. Noise inside the supplemental laboratory shall not exceed MIL-STD-1474, category F, with the exception of the utility compartment which will not exceed category E, when power generation and environmental control equipment is operating under full load and laboratory equipment and accessories are operating under typically encountered conditions.

3.20 Threaded parts. All screw threads shall be in accordance with ANSI Y14.6.

3.21 Hazardous materials. Provisions for the transportation of chemicals, compressed gasses, and other hazardous materials shall comply with the Department of Transportation Regulation 49CFR171-190 (see 6.4).

3.21.1 Safety engineering. The equipment design criteria shall be in accordance with Title 29, Code of Federal Regulations, Chapter XVII, Part 1910 "Occupational Safety and Health Standards"; in accordance with MIL-STD-454, requirement 1 and 3.18.

3.21.2 Chemical safety. Material safety data sheets (OSHA Form 20 or DD 1813) in accordance with FED-STD-313 shall be provided for each chemical to be delivered with the system and at the request of the contracting officer for chemicals associated with the system (see 6.9).

3.22 Cushion material. The cushion material for cabinets, drawers, and shelves shall conform to PPP-C-1752, type I, class 2. Color shall be white.

3.22.1 Installation. The cushion material shall be installed in all cabinets, cabinet drawers, and on all cabinet shelves except the overhead bookshelves, in a form corresponding to the interior of same. The depth of the cushion material shall be sufficient to protect stored equipment from impact and vibration damage. The material shall be altered in such fashion as required to conform to the profile of the stored equipment.

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3.23 Transportability. Provisions shall be made within the supplemental laboratory to secure all equipment, supplies, chemicals, books, binders and manuals. When transported as specified herein, no damage, dislocation, breakage, cracking, or leaking shall be evident. The supplemental laboratory, fully equipped (meaning all equipment, furnishings, and supplies installed and secured for transport), shall be transportable worldwide by the highway, water, and air modes. Transportability criteria are defined in MIL-STD-1366. Vehicles, semitrailer vans, and all other materials shall not be loaded beyond their rated capacities at any time. All slinging/tiedown provisions shall be labeled "LIFT", "TIEDOWN" or "LIFT TIEDOWN", as appropriate, in 1-inch (2.54 cm) high letters.

3.23.1 Highway. The supplemental laboratory, fully equipped, shall sustain over-the-road, across curbing, and unimproved roadway transport by 5-ton truck-tractors, military design M932 or equal. When towed by the M932 or equal under any of the conditions given in 3.23 and MIL-S-46756, 3.5, the supplemental laboratory shall experience no interference with the M932 or equal tractor.

3.23.2 Tiedown provisions. The tiedown provisions for the supplemental lab and for its bogie axle, when removed for air transport, shall conform to MIL-STD-209, class 2 or 3, type IV equipment, and to MIL-STD-1791 for equipment restraint criteria. The tiedown provisions shall satisfactorily complete the pull testing as specified without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components.

3.23.3 Slinging provisions. The slinging provisions for the supplemental lab and its bogie axle, when removed for air transport, shall conform to MIL-STD-209, class 1 or 3, type II. The provisions shall enable the complete laboratory to be lifted in the normal operating position. Any structure of an item (such as the corners where the top and side panels of the semitrailer van meet) that creates interference with a lifting sling will have the strength to withstand contact with the sling during pull testing without permanent deformation to any part of the item, and the contact will not adversely affect the sling. Slinging provisions may also be used as tiedown provisions when such provisions meet the requirements specified in 3.23.2.

3.23.4 Water. The supplemental laboratory, without disassembly, shall be capable of transport aboard general breakbulk (cargo), roll-on/roll-off (RORO), and barge-carrying (LASH and SEABEE) ships. The supplemental laboratory, when towed by the M932 or equal, shall be able to approach, negotiate, and depart a fifteen degree angle ramp without loss of tire to road surface contact.

3.23.5 Air. The supplemental laboratory, fully equipped, shall be transportable in a reduced configuration by C-141 or larger aircraft. All items removed for transport to include the tandem bogie axles, stowage compartments, and wastewater tank if necessary, shall be capable of being transported with their semitrailer van. The supplemental laboratory, fully

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equipped, shall be RORO transportable aboard C-5 aircraft without disassembly. Criteria for transport in U.S. Air Force cargo aircraft are defined in MIL-STD-1791.

3.24 Vehicle weight classification. The contractor shall complete the Government-furnished vehicle weight classification forms and apply the vehicle weight classification number(s) to the supplemental laboratory as specified (see 6.7).

3.25 Treatment and painting. The portions of the supplemental laboratory normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2), the supplemental laboratory shall then be overcoated in accordance with Government-furnished camouflage patterns with MIL-C-46168 or MIL-C-53039, top coat of the colors specified in camouflage patterns.

3.26 Finish. Cabinets and drawers containing material and apparatus shall be one color. Walls, bulkheads, and ceiling shall be a uniform, low-luster color (off-white, #27880). An epoxy modified, low-luster (semigloss) enamel paint shall be used. Stainless steel surfaces shall not be painted.

3.27 Identification marking. The supplemental laboratory and its integral parts shall be identified in accordance with MIL-STD-130. The marking shall be applied to the laboratory on plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous location.

3.28 Shipping data plate. The shipping data plate shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material and, in addition, shall show the silhouette of the supplemental laboratory in transport position indicating the center of gravity and the location and capacity of the lifting and tiedown attachments. The plate shall be attached by screws, bolts, or rivets in a conspicuous location.

3.29 Instruction plates. Each supplemental laboratory shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in the deployment, operation, or servicing of the laboratory in accordance with MIL-STD-1472. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. Plate shall be attached by screws, bolts, or rivets in a conspicuous location.

3.30 Government-furnished property. When specified (see 6.2), the following property in the quantities indicated will be furnished by the Government (see 6.5):

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Item no.	Description	Identification	Qty
1	Semitrailer, Van: M822E1	2330-01-050-8810	1
2	Air Conditioner	4120-00-168-1781	4
3	ASTM Standards, Petroleum Products and Lubricants, Vol. 05.01	(FSCM 81346)	1
4	ASTM Standards, Petroleum Products and Lubricants, Vol. 05.02	(FSCM 81346)	1
5	ASTM Standards, Petroleum Products and Lubricants, Vol. 05.03	(FSCM 81346)	1
6	Chemical Engineers Handbook	7610-00-250-6633	1
7	Chemistry and Physics Handbook	7610-00-233-9577	1
8	ASTM Viscosity Index	STP-168 or D 539	1
9	ASTM Viscosity Tables	STP-43-B	1
10	Chemical Index		1
11	Chemical Technology of Petroleum and Its Products		1
12	FED-STD-791	(FSCM 06542)	1
13	National Petroleum Correction Tables		1
14	Significance of ASTM Tests for Petroleum Products		1
15	Environmental Protection and Enhancement	AR 200-1	1
16		AR 310-1	1
17		AR 310-49	1
18		AR 340-1	1
19		AR 340-2	1
20		AR 340-18-1	1
21		AR 420-90	1
22		AR 623-1	1
23	Overseas Laboratories for Support of Quality Surveillance on Petroleum Products	AR 700-36	1
24	Procurement Quality Assurance	AR 702-4	1
25	Coal and Petroleum Products Supply and Management Activities	AR 703-1	1
26	Materiel Management for Using Units, Support Units, and Installations	AR 710-2	1
27	Storage and Supply Activity	AR 740-1	1
28	Petroleum Terminal and Pipeline Operations	FM 10-18	1
29		FM 10-19	1
30	Organizational Maintenance: Military Petroleum Pipelines, Tanks, and Related Equipment	FM 10-20	1
31	Manual, Operating and Maintenance, for the Laboratory, Petroleum, Semitrailer Mounted	FM 10-21	1

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Item no.	Description	Identification	Qty
32	Petroleum Supply in Theaters of Operation	FM 10-67	1
33	Aircraft Refueling	FM 10-68	1
34	Petroleum Supply Point Equipment and Operation	FM 10-69	1
35	Inspecting and Testing Petroleum Products	FM 10-70	1
36	Petroleum Tank Vehicle Operations	FM 10-71	1
37	Petroleum Surveillance: Laboratories and Kits	FM 10-72	1
38	How to Prepare and Conduct Military Training	FM 21-6	1
39	First Aid for Soldiers	FM 21-11	1
40		FM 21-20	1
41	Military Symbols	FM 21-30	1
42	NBC Defense	FM 21-40	1
43		FM 21-41	1
44		FM 21-48	1
45		FM 35-20	1
46	Symbols for the Identification of Fuels, Lubricants and Allied Products - NATO	TB 34-9-25	1
47	Calibration and Repair Requirements for the Maintenance of Army Materiel	TB 43-180	1
48	Recommended and Alternate Grade Fuels, Engine and Transmission Oils for Army Aircraft Engine Combinations	TB 55-9150-200-25	1
49	Procedures for the Management of of Petroleum Products	DOD 4140.25M	1
50		DOD 4155.1	1
51	Reference List of Specifications and Standards for Petroleum and Related Products	DFSCH 4120.1	1 1
52	Chemical, Biological, and Radiological Decontamination	TM 3-220	1
53	Firefighting and Rescue Procedures in Theaters of Operation	TM 5-315	1
54	Repairs and Utilities: Fire Protection Equipment and Appliances; Inspections, Operations, and Preventative Maintenance	TM 5-687	1
55	Manual, Air Conditioner Parts List	TM 5-4120-295-25	1
56	Manual, Air Conditioner Operation and Maintenance	TM 5-4120-295-15	1
57	Manual, Petroleum Handling Equipment and Operation	TM 10-1101	1

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Item no.	Description	Identification	Qty
58	Operator, Organization, Direct and General Support Maintenance Products	TM 9-2330-271-14	1
59	Identification Methods for Bulk Product Systems	MIL-STD-161	1
60	Packaging, Packing and Marking of Petroleum and Related Products	MIL-STD-290	1
61	Frequency for Identification and Cleaning of Petroleum Fuel Operating and Storage Tanks	MIL-STD-457	1
62	Petroleum Laboratory Calibration Systems	MIL-STD-978	1
63	ACS Chemicals	MIL-STD-1218	1
64	Quality Surveillance Handbook for Fuels, Lubricants and Related Products	MIL-HDBK-200	1
65	Military Standardization Handbook, Petroleum Operations	MIL-HDBK-201	1
66	Military Standardization Handbook, Logistics Data for Petroleum Planning	MIL-HDBK-210	1
67	Technical Methods of Analysis		1
68	Pattern Drawings	Camouflage Patterns	As reqd
69	Sign Kit	MIL-S-40626	1
70	Distillate Fuel Contamination, Storage, and Handling	ASTM STP 1005	1
71	Guide for the Selection of Lubricants Functional Fluids, Preservatives, and Specialty Products for Use in Ground Equipment Systems.	MIL-HDBK-113	1
72	Mobility Fuels Use Handbook	MIL-HDBK-114	1

Note: Items 3, 4, 5 and 10 through 67 do not possess a complete NSN as of the date of this publication.

3.31 Government-loaned property. The following property in the quantity indicated will be loaned by the Government (see 6.6).

Item No.	Description	Quantity
1	M818 truck tractor	As required

3.32 Workmanship.

3.32.1 Fabrication. Metal used in the fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of

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the metal used to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape.

3.32.2 Welding. The surfaces of parts to be welded shall be free from rust, oxides, scale, paint, grease, or other foreign matter. Weldments shall be of homogeneous appearance and form without spattering or other irregularities, complying with the size, configuration, and other dimensional requirements to develop the full strength of the parts joined by the welds. No cracks of any type are permitted. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.32.3 Welders. Before assigning any welder or welding operator to manual welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code - Steel
AWS D1.2, Structural Welding Code - Aluminum
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualification

Contractors who only make horizontal welds need not qualify welders for "all position welding." The contractor is responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes. In the event of evidence of poor welds, the Government reserves the right to require recertification of any welder or welding operator. The test results shall be made available for review by the contracting officer or the contracting officer's representative.

3.32.4 Welding practices.

a. Tack welds shall be subject to the same quality requirements as final welds.
b. Work shall be positioned for horizontal welding whenever practicable.
c. Procedures and sequences shall be such that distortion and shrinkage will be held to a minimum. When straightening is required, caution shall be exercised to ensure that the straightening process does not weaken the part, the assembly of which it is a component, or the supplemental laboratory.
d. Preheat of materials being welded and maximum interpass temperatures during welding shall be in accordance with the suppliers' recommendations.

3.32.5 Weld quality.

a. The contractor shall be responsible for determining the inspections needed to ensure when the welded parts are assembled together to make

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the supplemental laboratory, the laboratory shall conform to the inspection requirements specified in section 4.

b. All weldments shall be free of slag, flux, weld spatter, and other impurities detrimental to both the appearance and strength of the weldment.

c. Undercut in weldments shall be held to a minimum, but shall not be more than 0.03125 inch deep or 5 percent of the thickness of the thinner material, whichever is less.

d. Welds shall be free from all cracks in adjacent base material and weld metal, and from other discontinuities such as lack of fusion with base metal or between weld passes, incomplete penetration or lack of prescribed fit.

e. The weld quality shall be in accordance with AWS D1.1 or AWS D1.2 as appropriate.

3.32.6 Brazing. Brazing of steels, copper alloys and nickel alloys shall conform to MIL-B-7883.

3.32.7 Bolted connections. Bolt holes shall be accurately formed and shall have the burrs removed. Washers and lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall have full thread engagement. All threads shall be in accordance with FED-STD-H28.

3.32.8 Rivet connections. Rivets and installation thereof shall be in accordance with drawing 13217E1061.

3.32.9 Machine work. Tolerances and gauges for metal fits shall conform to the standards of the industry.

3.33 Environmental.

3.33.1 Operating temperature. The supplemental laboratory shall operate in ambient air temperatures ranging from -25 °F to 120 °F. Special environmental equipment shall be supplied as required.

3.33.2 Storage temperature. The supplemental laboratory shall not be damaged by storage at ambient temperatures from -28 °F to 160 °F.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the

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specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility of compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual; nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.7).

4.3 First article.

4.3.1 Examination. The first article supplemental laboratory shall be examined for the defects specified in table I. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article supplemental laboratory shall be tested as specified in 4.5.1 through 4.5.10. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each supplemental laboratory shall be examined for the defects specified in table I. Presence of one or more defects shall be cause for rejection.

TABLE I. Examination schedule

Number	Defects	Requirement paragraph
101.	Assembly incomplete or incorrect.	3.1
102.	Materials not as specified.	3.3

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TABLE I. Examination schedule - continued.

Number	Defects	Requirement paragraph
103.	Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.	3.3.1
104.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.2
105.	Contractor does not have documentation available for identification of material, material finishes, or treatment.	3.3.3
106.	Used, rebuilt or remanufactured components, pieces, or parts incorporated in the supplemental laboratory.	3.3.4
107.	Design detail not as specified.	3.4
108.	Semitrailer modifications incomplete or incorrect.	3.4.2
109.	Cabinetry not as specified.	3.4.2.5
110.	Parts and components missing or not as specified.	3.5
111.	Laboratory purging design not as specified.	3.6
112.	Environmental parameters not as specified.	3.7
113.	Ambient temperature not as specified.	3.7.1
114.	Air supply not as specified.	3.7.2
115.	Air conditioning units not as specified.	3.7.3
116.	Ducting not as specified.	3.7.4
117.	Heating not as specified.	3.7.5
118.	Environmental controls not as specified.	3.7.6
119.	Utility systems not as specified.	3.8
120.	Water system not as specified.	3.8.1
121.	Water system color code marking not as specified.	3.8.4
122.	Electrically operated utility and test equipment and systems not as specified.	3.9
123.	Primary (208-volt) electrical system not as specified.	3.9.1
124.	Secondary (110-volt) electrical system not as specified.	3.9.2
125.	Dedicated computer power circuit not as specified.	3.9.3
126.	Emergency electrical system not as specified.	3.9.5
127.	Instrument and support devices not as specified.	3.9.8
128.	115-Vac outlets not as specified.	3.9.9
129.	208-volt receptacles not as specified.	3.9.10
130.	Exhaust fan controls not as specified.	3.9.11

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TABLE I. Examination schedule - continued.

Number	Defects	Requirement paragraph
131.	Light switch and lamp location not as specified.	3.9.12
132.	Blackout lamps not as specified.	3.9.13
133.	Microswitch installation not as specified.	3.9.14
134.	Electrical wiring not as specified.	3.10
135.	Electrical circuitry, components, and connections not protected as specified.	3.10.3
136.	Safety equipment not as specified or missing.	3.11
137.	Placards and labels missing or incorrect.	3.13 & 3.14
138.	Receptacles, fixtures, and access panels not as specified.	3.15
139.	Utility lines not as specified.	3.16
140.	Illumination not as specified.	3.17
141.	Human Factors Engineering not as specified.	3.18
142.	Threaded parts not as specified.	3.20
143.	Hazardous material not installed as specified.	3.21
144.	Material safety data sheets missing or not prepared in accordance with FED-STD-313.	3.21.2
145.	Cushion material not as specified.	3.22
146.	Cushion material not installed as specified.	3.22.1
147.	Transportability not as specified.	3.23.1 thru 3.23.5
148.	Vehicle weight classification missing or not as specified.	3.24
149.	Treatment and painting not as specified.	3.25
150.	Paint color not as specified.	3.25
151.	Finish not as specified.	3.26
152.	Identification marking and identification plates not as specified or missing.	3.27
153.	Shipping data plate not as specified or missing.	3.28
154.	Instruction plates not as specified or missing.	3.29
155.	Workmanship not as specified.	3.32

4.5 Tests. Each supplemental laboratory shall be tested as specified in 4.5.1, 4.5.6, and 4.5.9. The Government shall pre-approve all test courses. Failure of any test shall be cause for rejection.

4.5.1 Test conditions. Tests shall be performed at the climatic conditions existing at the place of test. The supplemental laboratory shall operate as specified herein without maintenance other than the contractor's recommended

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normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test.

4.5.2 Road. The supplemental laboratory with all equipment, furnishings and supplies installed, shall be coupled with a Government-loaned M-52 truck tractor, NSN 2320-01-835-8326, or equal, and towed over a hard-surfaced road for a minimum of 200 miles at speeds up to at least 50 miles per hour. A circular test track shall not be used for this test. In addition, the combination shall negotiate at least 50 miles of trails (secondary unimproved roadway) at safe speeds (not less than 5 mph). Nonconformance to 3.23.1 shall constitute failure of this test.

4.5.3 Across curbing. The fully loaded and equipped supplemental laboratory shall be towed by the M818 truck tractor (or equal) over a section of concrete, asphalt, or solid wood curbing at not less than 5 miles per hour. The dimensions of the curbing shall be 6 inches, ±0.50 inch, high by 6 inches, ±0.50 inch, wide. The top corners of the curbing coming into contact with the wheels of the tractor and the laboratory shall be rounded to a 1-inch, ±0.125 inch, radius. The curbing shall be anchored to a hard-surfaced road and shall be stationary during the test. The full height of the curbing shall protrude above the surface of the road. All wheels of the tractor and laboratory shall pass over the curbing at a 45-degree, ±5 degrees, angle to the centerline of the curbing with the road side of the tractor and laboratory. This test shall be repeated passing all wheels of the tractor and laboratory over the curbing at a 45-degree, ±5 degrees, angle to the centerline of the curbing with the curb side of the tractor and laboratory. Nonconformance to 3.23.1 shall constitute failure of this test.

4.5.4 Hard braking test. The tractor shall be connected to the fully loaded and equipped laboratory. All equipment inside the unit and the overpack kits shall be mounted and anchored for transport. Two markers shall be placed on a level, straight section of ground 40 feet apart. The first marker shall be approached at a speed of 20 m.p.h. As the front of the tractor passes the first cone the brakes shall be applied and the tractor brought to a complete stop before reaching the second cone. Actual speed and stopping distance will then be recorded. The unit will then be inspected to ensure that no damage has occurred. Nonconformance to 3.23 shall constitute failure of this test.

4.5.5 Slinging and tiedown provision pull test. When specified (see 6.2), to determine conformance to 3.23.2 and 3.23.3, the slinging and tiedown provisions for the supplemental laboratory and for its bogie axle, when removed for air transport, shall be tested in accordance with MIL-STD-209. Test results that show that the provisions and the connecting structural members do not meet the requirements of MIL-STD-209 shall constitute failure of this test.

4.5.6 Pressure testing of utility system. The supplemental laboratory service utility systems shall be tested as indicated:

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System	Pressure	Medium	Duration*	Examination
Water	90 psig	Water	30	Visual
Drainage	Gravity	Water	10	Visual
Air	120 psig	Air	30	Soap solution
CO_2/O_2	Cylinder pressure	CO_2	30	Visual
Vacuum	13.5 M Pa	Negative pressure	30	Manometer

*Not less than time indicated in minutes

The above testing and examination shall be conducted for a total of two times. After the first test, the pressures shall be allowed to return to ambient before pressurizing for the second test. Any evidence of leakage shall constitute failure of this test.

4.5.7 Transport by water test. The supplemental laboratory shall be evaluated for conformance to 3.23.4. To determine conformance to 3.23.4, the supplemental laboratory, without disassembly, towed by the M932 or equal, shall approach, negotiate, and depart a fifteen degree angle ramp. Inability to be transported by breakbulk, RORO, and barge-carrying ships, or inability to approach, negotiate, and depart a fifteen degree ramp angle, when towed by the M932 or equal, without disassembly, undercarriage interference, and loss of tire to road surface contact shall constitute failure of this test.

4.5.8 Fixed wing aircraft loading test. The supplemental laboratory shall be inspected and calculations provided to show that it meets the criteria of MIL-STD-1791. Unless otherwise specified (see 6.2), the contractor will prepare for loading, load, and secure the supplemental laboratory in a C-130 aircraft in conformance with 3.23.5. Nonconformance to 3.23.5 or the criteria in MIL-STD-1791 shall constitute failure of this requirement.

4.5.9 Functional testing of electrically operated systems and equipment. The electrically operated equipment and electrical systems shall be electrically energized and operated in accordance with the laboratory technical manual.

a. Time-delay purging system.
b. Lighting blackout system.
c. Utility equipment.
d. Testing equipment.
e. 24-volt dc electrical system.
f. 115-volt ac electrical system.

Nonconformance to 3.9 shall constitute failure of this test.

4.5.10 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by

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MIL-STD-1474, figure 11. As a minimum, noise levels shall be measured when power generation and environmental control equipment is operating under full load and laboratory equipment and accessories are operating under typically encountered worst case conditions. MIL-STD-1474 contours shall be taken at not fewer than 12 equal arc increments; one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating or working position(s) shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 and 3.19 visions shall constitute failure of this test.

4.6 Fume hood face velocity test. Air velocities shall be measured by making a nine-point traverse (more for a large hood) with an anemometer or velometer in the face plane of the hood with the sash fully open. The average of the readings shall be at least 100 fpm, and the velocity at each point should not vary by more than 20 percent from the average. Nonconformance to 3.5.1 shall constitute failure of this test.

4.6.1 Smoke test for fume hood. Smoke candles or smoke ejectors shall be used to perform this test. Two tests shall be performed, one with the sash height at fully open position and one with sash at midway or midpoint. The candle in both tests shall be placed in the center of the fume hood approximately 4 inches from the front face. The smoke flow shall then be used to check: the vector of the air entering the hood face, back flows or leaks from the hood, turbulence within the hood work space, leaks in the hood, and how rapidly the hood cleared itself of smoke.

4.7 Environmental testing.

4.7.1 High temperature. The supplemental laboratory shall be tested in a high temperature environment in accordance with MIL-STD-810, method 501, procedures I and II. Temperature sensors shall be placed in all compartments (utility, main, computer). The sensors shall be placed at locations away from vents and any direct air flow. Test temperatures shall be as stated in 3.33. Test duration shall be seven cycles for procedures I and constant temperature exposure for II. The system shall be operated until all recorded temperatures have stabilized or a minimum of 4 hours. Interior operating temperatures must be between 50 °F and 75 °F for a minimum of 4 hours. Any damage, leakage or failure of the system to operate as required in this specification shall constitute failure of this test.

4.7.2 Low temperature. The supplemental laboratory shall be tested in a low temperature environment in accordance with MIL-STD-810, method 501, procedures I and II. Temperature sensors shall be located in all compartments (utility, main, computer room). Sensors shall be placed in strategic locations away from vents and any direct air flow. Test temperatures shall be as stated in 3.33. Interior operating temperatures must be maintained between 50 °F and 75 °F for a minimum of 4 hours. Test duration for storage is 4 hours and a minimum of 4 hours for operation until recorded temperatures stabilize. Any damage, leakage, or failure of the system to operate as required in this specification shall constitute failure of this test.

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4.8 Inspection of packaging. The packing and marking of each supplemental laboratory shall be examined for the following defects. Presence of one or more defect shall be cause for rejection.

156. The supplemental laboratory not preserved as specified for level A (see 5.1.1).
157. Drive belt tension not loosened as specified for level A (see 5.1.1.2).
158. Pulley grooves not coated with primer as specified for level A (see 5.1.1.2).
159. Ferrous metal surfaces requiring a preservative and not coated as specified for level A (see 5.1.1.2).
160. Specific types of items not preserved as specified for level A (see 5.1.1.1).
161. Fluorescent lamps not secured as specified for level A or C (see 5.1.1.2).
162. Exterior openings into air conditioner not closed as specified for level A (see 5.1.1.3).
163. Doors, blackout covers, and weather flaps not closed and secured as specified for level A or C (see 5.1.1.4).
164. Equipment, supplies, accessories, overpack boxes not positioned and secured as specified for level A or C (see 5.2).
165. Overpack boxes, if supplied, not as specified for level A or C (see 5.2).
166. Marking incomplete, illegible, incorrect, or missing for level A or C (see 5.3).

5. PACKAGING

5.1 Preservation. Preservation of the laboratory trailer and all components of the laboratory shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Trailer. The trailer shall be preserved in accordance with the level A requirements of MIL-V-62038 for level A towed equipment.

5.1.1.2 Laboratory components. All equipment and accessories shall be located and secured in their designated locations. Any components and accessories having electrical components such as motors, switches, etc., shall be preserved in accordance with MIL-P-116, method IIC. Any components which will not fit into designated cabinet space because of increased size due to preservation shall be packed in one or more boxes conforming to PPP-B-601, domestic type. Components shall be secured within the box(es) with blocking and bracing to prevent movement of contents. Exposed, unprotected ferrous metal surfaces of mounted equipment shall be coated with appropriate preservatives in accordance with MIL-P-116. All equipment driven by belts shall have the belt tension loosened and the pulley grooves coated with primer conforming to TT-P-664. Fluorescent lamps shall be secured in position by taping over the slot of the holders with 1-inch wide tape conforming to

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PPP-T-60, type IV. Manuals and books shall be preserved in accordance with MIL-P-116, method IC-3. The refrigerator door shall be propped and secured in the open position to prevent permanent set of the door gasket. Overpack boxes, if included, shall be secured inside the trailer in a manner which will prevent box movement in any direction.

5.1.1.3 Air conditioning unit. Exterior openings into the air conditioning unit not provided with covers shall be closed with waterproof barrier material conforming to PPP-B-1055, class H-5, secured in place with waterproof tape. A warning tag shall be prominently affixed to the air conditioning controls, and shall caution against operating the unit without all of the protective storage and transport material.

5.1.1.4 Door and exterior openings. The doors, blackout covers, and weather flaps shall be closed and secured. All inlets, outlets, joints around doors, covers, and flaps shall be sealed with minimum 2-inch wide tape conforming to PPP-T-60, type IV. Any openings too large to be sealed with tape shall be covered with waterproof barrier material secured in place with waterproof tape.

5.1.2 Level C. The trailer shall require no preservation. All equipment, supplies, accessories, and overpack boxes shall be positioned and secured as specified for level A. Fluorescent lamps shall be secured in place with the tape as specified for level A. Doors, blackout covers, and weather flaps shall be closed and secured. Additional preservation shall be applied, as deemed necessary by the contractor, to afford protection against deterioration and damage from the contractor to the initial destination.

5.2 Packing. Packing shall be level A or C, as specified (see 6.2).

5.2.1 Level A. The supplemental laboratory preserved as specified in 5.1 shall not require additional packing.

5.2.2 Level C. The supplemental laboratory preserved as specified in 5.1 shall not require additional packing, except doors shall be equipped with tamperproof seals to assure carrier acceptance at the lowest ratings of the carrier.

5.3 Marking. In addition to any caution and deprocessing marking, interior packs and units shall be marked in accordance with MIL-STD-129 and as specified herein. Special marking requirements applicable to unpacked vehicles shall apply. Tool, spare parts, and overpack boxes shall contain a list of items contained therein.

5.3.1 Humping precaution. Large signs containing the words "DO NOT HUMP" shall be secured in conspicuous places on each side of the supplemental laboratory.

5.4 Shipping precautions.

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5.4.1 Highway towaway shipment. Prior to highway shipment, no processing of the supplemental laboratory's running gear shall be performed that would impair necessary operation or safety during transit. Towaway rules for shipment by drive-away shall be in accordance with the DOT Motor Carrier Safety regulations, 49CFR393. When vehicles are to be placed in storage or shipped overseas, processing of the trailer unit's running gear not accomplished prior to highway shipment shall be performed at an intermediate designation.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The supplemental laboratory and sustaining equipment covered by this specification are to be used by Armed Forces for testing petroleum and petroleum products under tactical military operations in conjunction with the Semitrailer-mounted Petroleum Laboratory to form the Modular Base Petroleum Laboratory (MBPL).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Time frame required for submission of first article model (see 3.2).
d. When a semitrailer is required (see 3.4.1).
e. When items of equipment shall be furnished by the contractor (see 3.5).
f. Weight classification numbers and instructions required (see 3.24 and 6.7).
g. When overcoat in accordance with camouflage patterns is not required (see 3.25).
h. When Government-furnished property is required (see 3.30 and 6.5).
i. When MIL-STD-209 provision pull testing is required for Government furnished property (see 4.5.5).
j. When an aircraft test loading is not required for a semitrailer (see 4.5.8).
k. Level of preservation and packing required (see 5.1 and 5.2).
l. When a transportability report is required (see 6.11).

6.3 First article. When a first article inspection is required, the item should be a first-produced model. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering

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a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Hazardous materials. Hazardous materials are listed in DoD Regulation 49CFR171 (see 3.21).

6.5 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.30, as applicable.

6.6 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.31.

6.7 Vehicle weight classification. The contracting officer should arrange to furnish vehicle classification forms to the contractor and request the return of the completed forms to the Government. The information thus obtained should be utilized to determine the weight classification number(s) for the laboratory. The contractor should then be requested by the contracting officer to apply the weight classification number on the front of the supplemental laboratory on the curb side. The use of a Government-furnished sign kit conforming to MIL-S-40626 should be utilized where applicable (see 3.24).

6.8 Definition. Where an item is specified by trade name and is followed by the words "or equal" or "or equivalent," it is intended to indicate a supplier satisfactory to the Government and a quality level that must be met or surpassed by any supplier of an "or equal" or "or equivalent" item. In order to be considered equal or equivalent, a piece or item must be dimensionally and functionally interchangeable with the piece or item it is to replace.

6.9 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed material safety data sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in appendix B of FED-STD-313.

6.10 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.11 Transportability report. When specified (see 6.2) the contracting officer should arrange for the contractor to furnish a transportability report

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in accordance with AR 70-47 and S.O.P. 70-38 for the supplemental laboratory 95 days prior to delivery of the first article supplemental laboratory.

6.12 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of compactors.

6.13 Subject term (key word) listing.

Analysis
Fuel
MBPL
Oil
Testing, Petroleum

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activities:
Army - MD

Project No. 6640-A611

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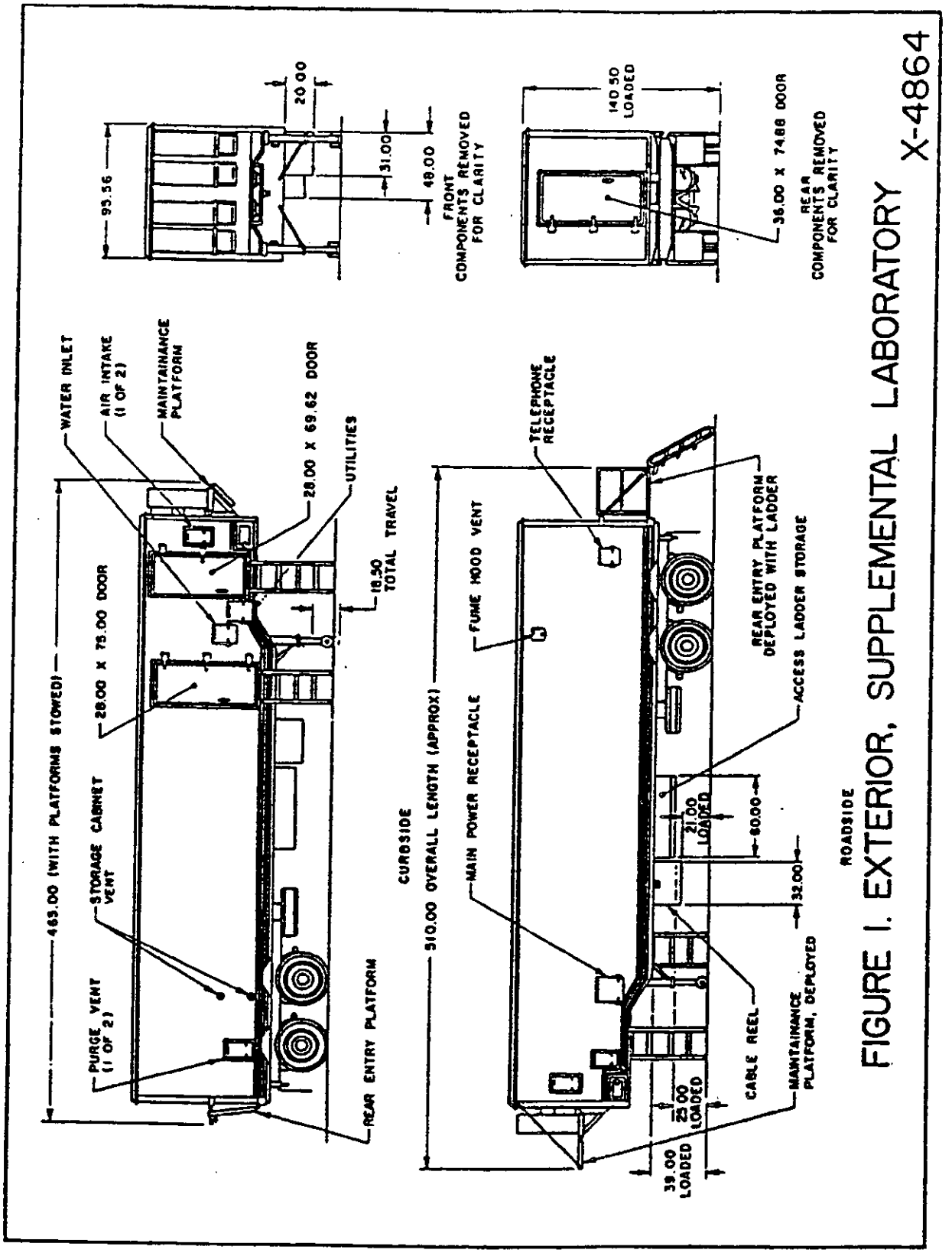


FIGURE I. EXTERIOR, SUPPLEMENTAL LABORATORY

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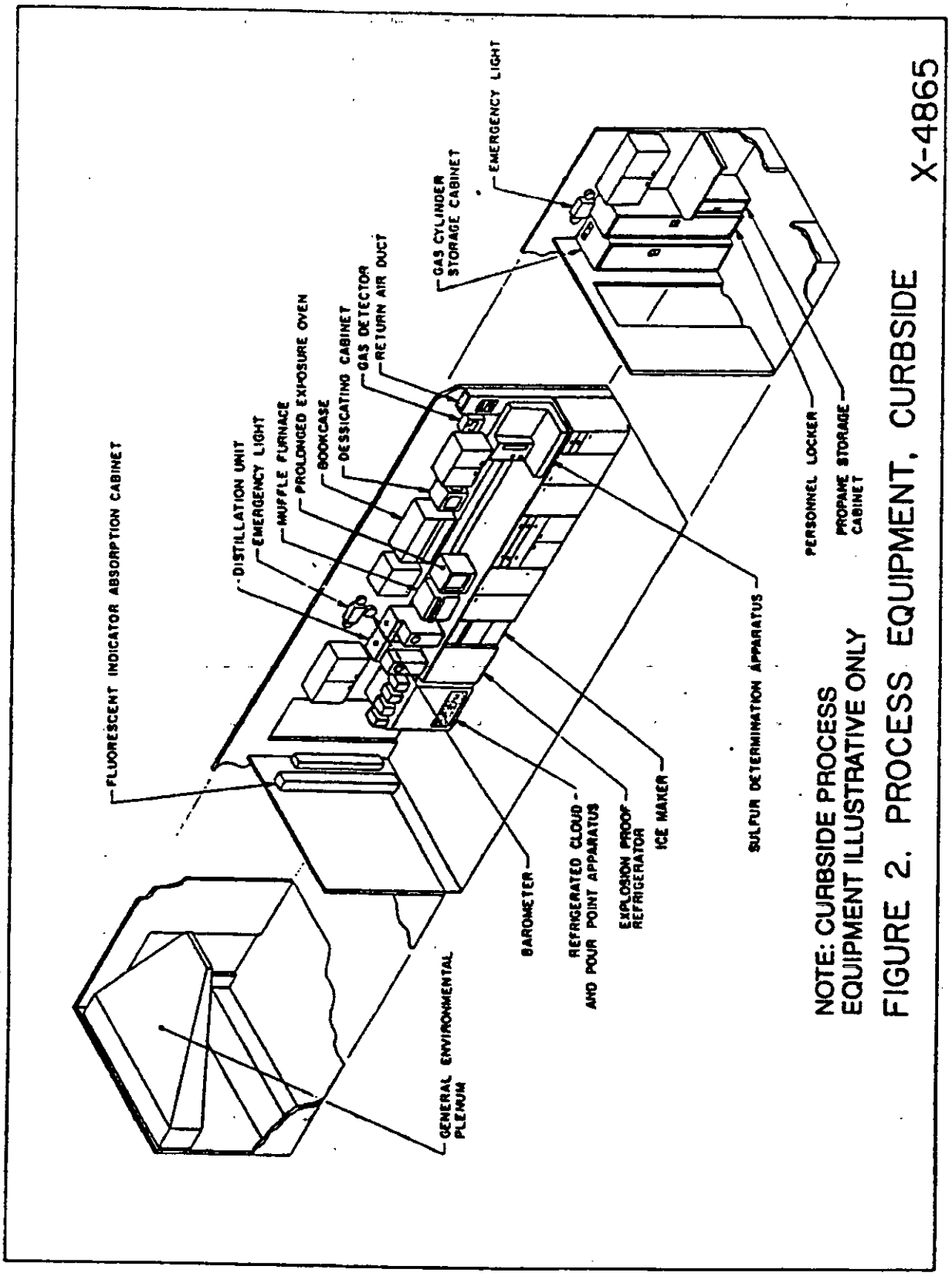
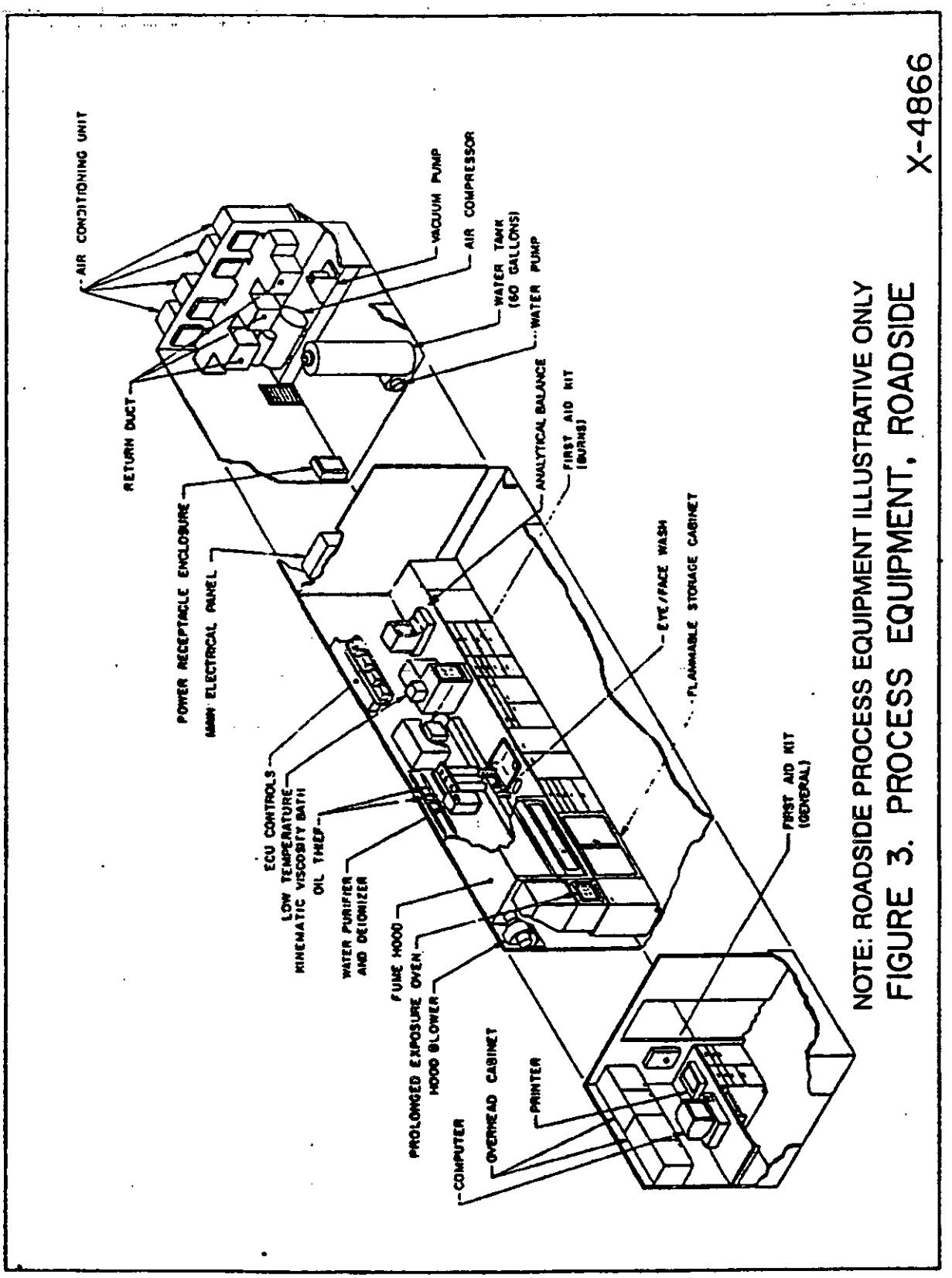


FIGURE 2. PROCESS EQUIPMENT, CURBSIDE

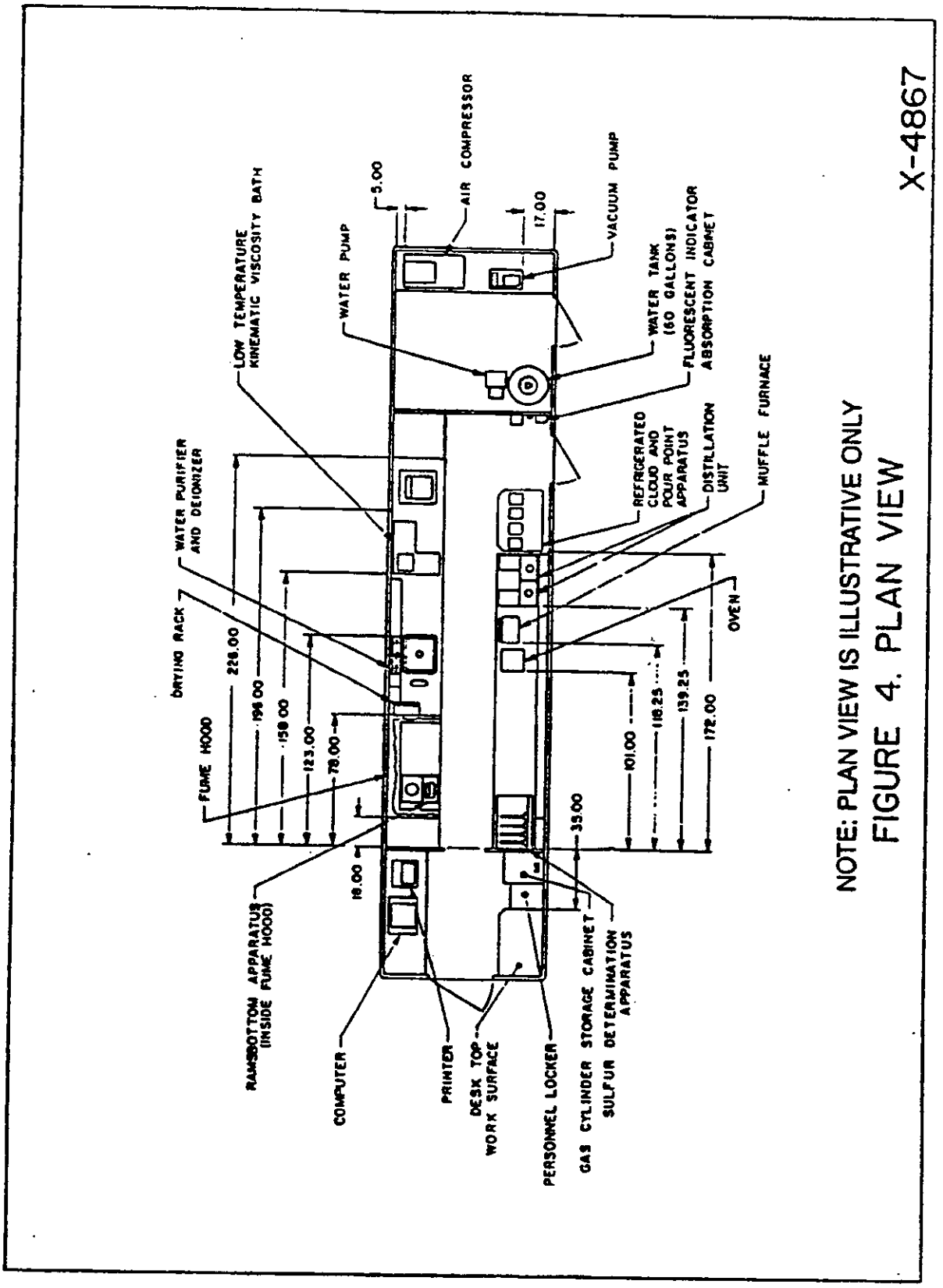
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NOTE: ROADSIDE PROCESS EQUIPMENT ILLUSTRATIVE ONLY

FIGURE 3. PROCESS EQUIPMENT, ROADSIDE

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-L-53111(ME)	91/09/12

3. DOCUMENT TITLE
Laboratory, Petroleum, Supplemental, Semitrailer Mounted

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(703) 664-5717	354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466	

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